LSTM vs. Transformer Encoder for Text Classification: A Controlled Comparison on AG News

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Abstract

We present a controlled comparison of a recurrent encoder (LSTM) and a self-attention encoder (Transformer) for AG News four-class topic classification, with both models trained from scratch. We hold the data, preprocessing, vocabulary, sequence length, and training budget fixed and vary only the encoder, so that any difference in behaviour can be attributed to the architecture rather than to confounding factors. Under identical conditions the Transformer reaches 0.910 test accuracy versus 0.833 for the LSTM at a comparable parameter count. Loss curves and a data-efficiency ablation show that the gap stems from generalization: the LSTM overfits and saturates at roughly 25% of the data, whereas the Transformer scales steadily. Removing positional encoding leaves accuracy unchanged, indicating that the task is essentially bag-of-words and that the gap arises from robustness, not word-order modelling. Token-importance analysis shows that the Transformer distributes evidence across several topical tokens while the LSTM can over-rely on a single token. The prior hypothesis that the Transformer is more data-hungry is rejected.

1. Introduction

Recurrent networks such as the LSTM [1] and self-attention networks such as the Transformer [3] are the two most widely used sequence encoders. This project compares them on AG News four-class news topic classification under identical conditions, with both models trained from scratch. Our research question is: when the data, preprocessing, vocabulary, sequence length, and training budget are all held fixed, how do the two architectures differ in performance, convergence, data efficiency, and error patterns? The goal is not to maximize accuracy itself but to explain the cause of the difference. A controlled, head-to-head comparison on the same task exposes the strengths and weaknesses of each architecture.

Class	Train	Test
World	26,991	1,900
Sports	26,966	1,900
Business	27,100	1,900
Sci/Tech	26,943	1,900
Total	**108,000**	**7,600**

Table 1. Per-class counts. Validation is 12,000 (10% held out from Train, seed 42). All classes are balanced.

2. Dataset

We use AG News from a single HuggingFace source, with four classes (0 World, 1 Sports, 2 Business, 3 Sci/Tech) [4]. The test set is reserved for final evaluation, and the validation set is formed by holding out 10% of the training set (seed 42). Table 1 summarizes the splits and the class distribution; all four classes are balanced.

Preprocessing is identical for both models. After word-level tokenization (lowercased), the vocabulary is built from the training set only (maximum size 20,000, minimum frequency 2, including padding and unknown tokens). Each sentence is truncated or padded to a maximum length of 128, and padding positions are ignored by masking. The training set contains 67,575 unique words; the 20,000 most frequent cover 97.3% of all token occurrences, so only about 2.7% of tokens map to the unknown token.

Label indexing is a common pitfall: the raw CSV uses labels 1–4 whereas the HuggingFace version uses 0–3, which is directly compatible with the 0-based index required by the cross-entropy loss. Using a single source avoids this confusion.

	LSTM	Transformer
Embed. dim	128	128
Encoder	bidirectional LSTM, 2 layers, hidden 128	2 layers, 4 heads, FFN 256, sinusoidal PE
Output dim	256 (bi)	128
Pool / Loss	masked mean / CE	masked mean / CE
Params	3,220,484	2,825,476

Table 2. Model configurations. Only the encoder differs; the embedding, pooling, and head are identical.

3. Models

Both models share the same skeleton: an embedding layer (dimension 128), an encoder, masked mean pooling, and a linear output layer (4 classes), trained with the cross-entropy loss. Only the encoder in the middle differs. This control is the core of the comparison and lets us attribute performance differences to the architecture. Table 2 lists the configurations. Because the embedding (about 2.56M parameters) dominates both models, the total parameter counts differ by only about 14% and are therefore comparable.

Implementation details: the padding token is fixed at index 0 and registered as the embedding padding index, so its vector is never trained. Pooling and self-attention ignore padding positions via masking. Both models use the same masked mean pooling, so that the pooling method is not a confounding factor.

4. Experiments

The training setup is the Adam optimizer [2] with learning rate 0.001, batch size 64, up to 8 epochs, dropout 0.1, and seed 42. Model selection uses the best checkpoint by validation performance only, and the test set is evaluated exactly once. Training runs on Apple MPS.

The main comparison trains both models with the setup above. The ablation measures data efficiency by reducing the training set to 5%, 25%, 50%, and 100%. The vocabulary is built once on the full training set and reused unchanged at every fraction; subsets are stratified by class with a fixed seed; the validation and test sets are always used in full. The evaluation metrics are accuracy, macro F1-score, training/validation loss curves, and confusion matrices. For a fair comparison the split, tokenizer, vocabulary, maximum length, pooling, optimizer, learning rate, batch size, number of epochs, and seed are all identical across the two models; the only variable is the encoder.

Model	Test accuracy	Test macro F1
LSTM	0.833	0.835
Transformer	**0.910**	**0.909**

Table 3. Main results under identical conditions.

5. Results

Table 3 shows that the Transformer leads by about 7.7 percentage points at a comparable parameter count.

The learning curves are direct evidence of the gap (Fig. 1). For the LSTM, the training loss falls to 0.035 while the validation loss rises from 0.54 to about 2.0, a textbook case of overfitting. The Transformer keeps its validation loss flat (about 0.28 to 0.33) with almost no overfitting and converges quickly, already reaching 88% accuracy after the first epoch. Because the best checkpoint by validation is saved before overfitting (epoch 2), the LSTM still secures its final score.

The confusion patterns (Fig. 2) are as follows. In the LSTM, Business is mostly misclassified as Sci/Tech (258 cases) and Sci/Tech mostly as Business (154 cases), so the mutual confusion between these two classes is largest. World and Sports have similar error distributions, both misclassified mainly as Business and Sci/Tech; as a result the LSTM makes substantial errors in every class, spread across the board. In the Transformer, errors concentrate almost entirely on the Business/Sci-Tech boundary (Business as Sci/Tech 147 cases, Sci/Tech as Business 138 cases), while Sports is near-perfect (38 errors) and World is clean.

The shared trait is that both models confuse Business and Sci/Tech the most, because the two classes share business, technology, and market vocabulary; the cause is semantic overlap, not sequence length. Per class, the Transformer is especially strong on Sports and World and weak only on Business and Sci/Tech. The LSTM is not as strong as the Transformer even on Sports and World, making non-trivial errors even where the vocabulary is distinctive.

The data-efficiency ablation (Fig. 3, Tab. 4) is revealing. The Transformer improves monotonically as data grows. The LSTM peaks at 25% (0.846) and then plateaus; adding more data does not help because of overfitting. In other words, the LSTM saturates at about the 25% point and cannot exploit additional data, while the Transformer scales steadily with data. Because the Transformer leads at every fraction, the prior hypothesis that the Transformer requires more training data (i.e., is less data-efficient) is rejected for this task. AG News is a relatively easy task, so the Transformer generalizes well even at the 5,400-example

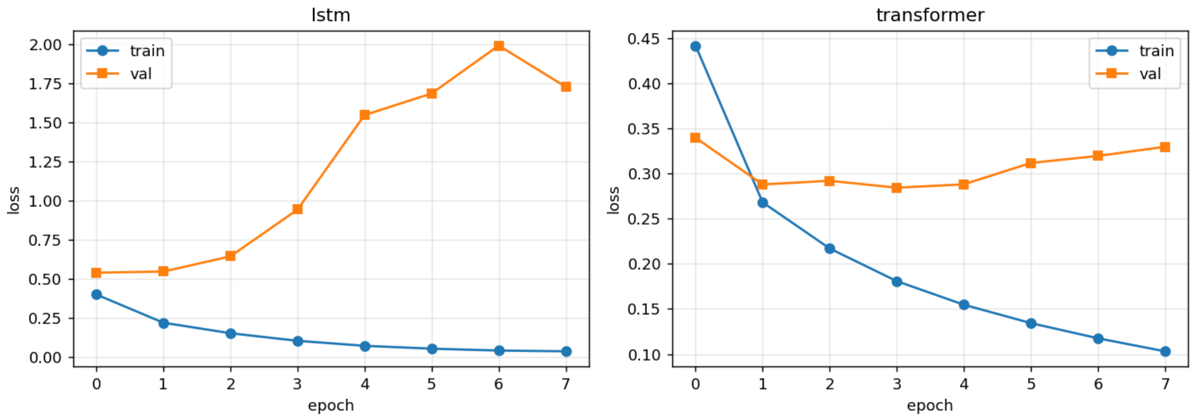

Figure 1. Training and validation loss curves (left: LSTM, right: Transformer). The LSTM overfits (validation loss rises to about 2.0), while the Transformer keeps validation loss flat.

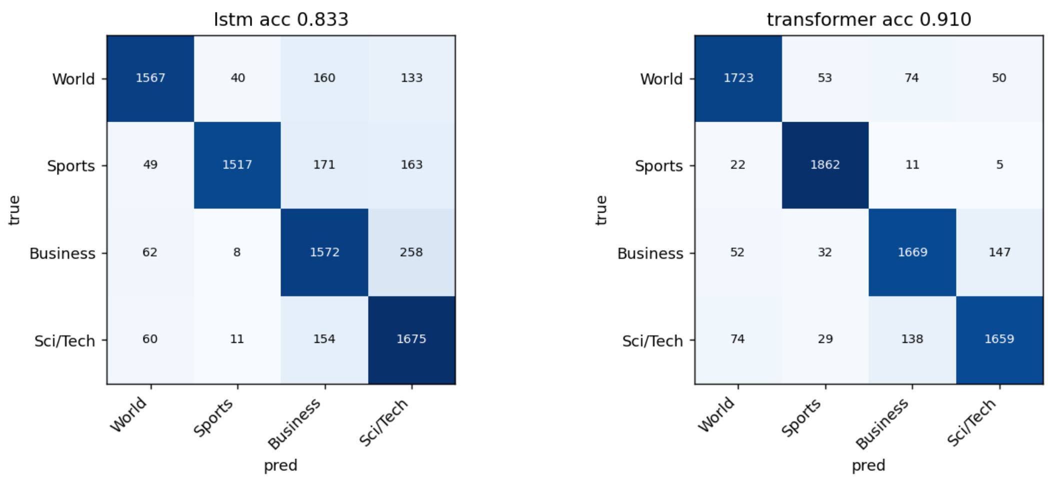

Figure 2. Confusion matrices (row: true label, column: predicted label). Both models confuse Business and Sci/Tech most; the LSTM additionally errs across all classes.

Training data	LSTM	Transformer	Gap
5% (5,399)	0.727	0.798	+0.071
25% (26,998)	0.846	0.875	+0.029
50% (53,999)	0.833	0.897	+0.064
100% (108,000)	0.833	**0.910**	+0.077

Table 4. Data-efficiency ablation: test accuracy vs. training-set size.

scale.

Training efficiency also differed. On Apple MPS, the per-epoch training time was about 47 s for the Transformer versus about 56 s for the LSTM, because self-attention parallelizes computation across positions whereas the LSTM processes tokens sequentially. The Transformer thus led in both accuracy and training speed.

Overall, the performance gap stems from generalization and overfitting rather than from sequence-order modelling (which, as the next section shows, this task barely uses).

6. Failure Analysis

By category, both models are wrong on 428 cases, the LSTM alone on 841, and the Transformer alone on 259. The LSTM's solo errors are about 3.2× more frequent, which supports the accuracy gap. Of the cases both models miss, 61% have a true label of Business or Sci/Tech; these are ambiguous boundary cases that reflect class overlap in the task rather than a model

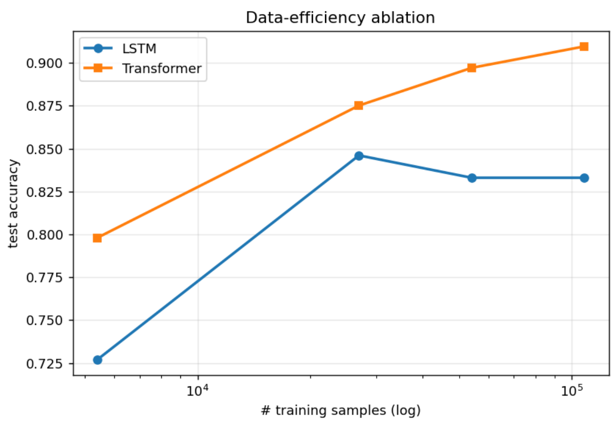

Figure 3. Data-efficiency ablation. Test accuracy versus training-set size. The LSTM saturates at 25%; the Transformer keeps improving.

defect (Tab. 5).

Confidence patterns also differ. Shared errors often come with high confidence: on the Intel example both models were wrong with confidence above 0.98, an overconfident boundary error. LSTM-only errors, by contrast, often carry low confidence (0.50–0.59), indicating misclassification under the model's own uncertainty and suggesting a lack of robustness. In short, the two models fail in different ways.

7. Extension: Transformer Mechanism Analysis

We carried out all three planned mechanism analyses; only the positional-encoding experiment requires extra training, while the others reuse the trained models.

First, a positional-encoding on/off experiment. Retraining the Transformer without positional encoding leaves test accuracy essentially unchanged (0.910 to 0.911, a difference of -0.001). Without positional encoding, self-attention is a set encoder with no order information [3]; identical accuracy therefore indicates that AG News is bag-of-words in nature, where token occurrence matters more than word order, and that the order-injecting component is unnecessary for this task.

Second and third, token-importance analyses via gradient saliency and attention weights. Fig. 4 compares which tokens each model relies on for the same Sports article. The LSTM assigns the largest importance to "giddy," a word unrelated to the topic, and misclassifies the article as Sci/Tech. The Transformer spreads importance across topical tokens such as "phelps," "seconds," "medley," and "400," with gradient saliency and attention weights showing the same trend, and classifies the article correctly as Sports. In other words, the Transformer distributes its evidence across several topical tokens, whereas the LSTM can over-rely on a single token; this reveals a token-level cause of the accuracy gap. We note that gradient saliency and attention weights do not coincide exactly, so attention weights should not be equated directly with explanation.

8. Conclusion

- Under identical conditions the Transformer is clearly superior to the LSTM (0.910 vs. 0.833 test accuracy).
- The cause is a difference in generalization. The LSTM overfits severely and saturates at roughly 25% of the data, failing to exploit additional data, whereas the Transformer scales steadily.
- The task is essentially bag-of-words: removing positional encoding does not affect accuracy, so the gap arises from robustness and generalization rather than word-order modelling.
- The prior hypothesis that the Transformer needs more training data is rejected; the Transformer leads at every data fraction.
- Limitations and future work. Results are based on a single seed and should be re-confirmed across seeds. Stronger regularization (e.g., dropout) for the LSTM may narrow the gap. The effect of positional encoding should be re-examined on order-sensitive tasks (sentiment analysis, natural language inference), and a word-order-shuffling ablation could compare the two architectures' order sensitivity more directly.

References

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[2] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In International Conference on Learning Representations (ICLR), 2015. 2

[3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS), 2017. 1, 4

[4] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In Advances in Neural Information Processing Systems (NeurIPS), 2015. 1

Article (summary)	True	LSTM	Transf.	Likely cause
Some People Not Eligible to Get in on Google IPO	Sci/Tech	Business	Business	Corporate IPO blurs the finance/technology boundary
Intel to delay product aimed for high-definition TVs	Business	Sci/Tech	Sci/Tech	Tech vocabulary but a business-impact label; both wrong with high confidence
Teenage T. rex's monster growth	Sci/Tech	Business	Sci/Tech	Clearly science; only the LSTM errs, with low confidence
IBM to hire even more new workers	Sci/Tech	Business	Sci/Tech	Hiring story; boundary is ambiguous; only the Transformer is correct
Prediction Unit Helps Forecast Wildfires	Sci/Tech	Sci/Tech	Sports	A rare Transformer error

Table 5. Representative misclassifications and their likely causes.

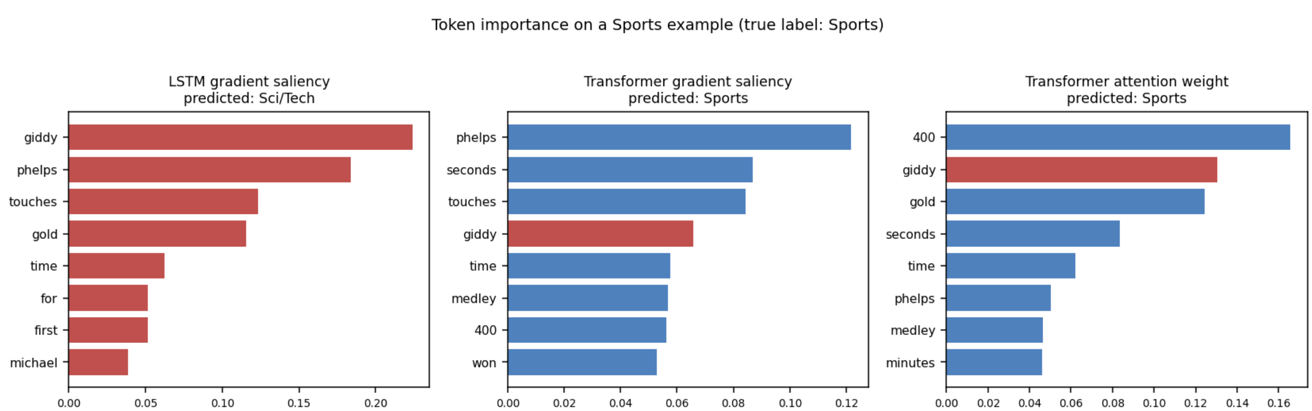

Figure 4. Token importance on a Sports example (true label: Sports). Left: LSTM gradient saliency (misclassified as Sci/Tech). Centre and right: Transformer gradient saliency and attention weight (both correct).